

Corrected $\mathcal{O}(v^2)$ Dispersion Relation via WKB Approximation

1. Derivation of the Governing Equations to $\mathcal{O}(v^2)$

Starting from the generalized wave equation, we expand the terms up to order $\mathcal{O}(v^2)$, utilizing the relation $\beta = \frac{\alpha v}{2}$.

Domain $x > 0$

The equation is expanded as:

$$k_z^2 - \partial^2 - \omega^2 - \frac{\alpha^2 v^2}{4} x^4 = \frac{\alpha^2 v}{\omega(\omega - k_z v)} \left(v - \frac{k_z + vx^2 \frac{\alpha}{2}}{\omega + v^2 x^2 \frac{\alpha}{2}} \right) \quad (1)$$

Applying the Taylor expansions for the denominators:

$$\frac{1}{\omega - k_z v} \approx \frac{1}{\omega} \left(1 + \frac{k_z v}{\omega} \right), \quad \frac{1}{\omega + \beta v x^2} \approx \frac{1}{\omega} \left(1 - \frac{\alpha v^2 x^2}{2\omega} \right) \quad (2)$$

Substituting these into the right-hand side yields:

$$\text{RHS} = \frac{\alpha^2 v}{\omega^2} \left(1 + \frac{k_z v}{\omega} \right) \left[v - \left(k_z + \frac{\alpha v x^2}{2} \right) \left(\frac{1}{\omega} - \frac{\alpha v^2 x^2}{2\omega^2} \right) \right] \quad (3)$$

Expanding and keeping terms up to $\mathcal{O}(v^2)$:

$$\text{RHS} = \frac{\alpha^2 v^2}{\omega} - \frac{k_z \alpha^2 v}{\omega^2} \left(1 + \frac{k_z v}{\omega} \right) - \frac{\alpha^3 v^2}{2\omega^2} x^2 \quad (4)$$

Rearranging into standard operator form gives:

$$\partial^2 a_z + C_1 x^2 a_z + C_2 x^4 a_z + E_+ a_z = (k_z^2 - \omega^2) a_z \quad (5)$$

where the coefficients are defined as:

$$C_1 = -\frac{\alpha^3 v^2}{2\omega^2}, \quad C_2 = \frac{\alpha^2 v^2}{4} \quad (6)$$

$$E_+ = \gamma_1 \left(\gamma_2 - \frac{k_z}{\omega} \right), \quad \text{with} \quad \gamma_1 = \frac{\alpha^2 v}{\omega}, \quad \gamma_2 = v \left(1 - \frac{k_z^2}{\omega^2} \right) \quad (7)$$

Domain $x < 0$

Following the equivalent expansion for the negative domain with the altered Doppler shift:

$$\partial^2 a_z + \omega^2 - k_z^2 + \frac{\alpha^2 v^2}{4} x^4 = \frac{\alpha^2 v^2}{\omega} + \frac{\alpha^2 k_z v}{\omega^2} \left(1 - \frac{k_z v}{\omega} \right) + \frac{\alpha^3 v^2}{2\omega^2} x^2 \quad (8)$$

Rearranging to match the format of the positive domain:

$$\partial^2 a_z + C_1 x^2 a_z + C_2 x^4 a_z - E_- a_z = (k_z^2 - \omega^2) a_z \quad (9)$$

where C_1 and C_2 are identical to the positive domain, and:

$$E_- = \gamma_1 \left(\gamma_2 + \frac{k_z}{\omega} \right) \quad (10)$$

2. The Effective Potentials

We consolidate the equations into the standard form $\partial_x^2 a_z + Q_\pm(x) a_z = 0$.

$$Q_+(x) = \omega^2 - k_z^2 + E_+ - |C_1|x^2 + C_2x^4 \quad (11)$$

$$Q_-(x) = \omega^2 - k_z^2 - E_- - |C_1|x^2 + C_2x^4 \quad (12)$$

Evaluating the constant terms using the definitions of E_+ and E_- :

$$Q_0^+ = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2} + \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2}\right) \quad (13)$$

$$Q_0^- = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2} - \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2}\right) \quad (14)$$

This reveals a continuous linear driving term $Q_{0,\text{mean}} = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2}$ and a discontinuous quadratic jump $\pm \delta = \pm \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2}\right)$.

1. The Physical Premise of the WKB Method

The WKB (Wentzel-Kramers-Brillouin) approximation is a technique used to find solutions to linear differential equations with spatially varying coefficients. We have cast our system into the standard Schrödinger-like form:

$$\partial_x^2 a_z + Q(x) a_z = 0 \quad (15)$$

where $Q(x)$ acts as the “effective momentum squared” of the wave.

If $Q(x) > 0$, the wave oscillates. If $Q(x) < 0$, the wave decays exponentially. The boundaries separating these regions are called **turning points** (x_T), where $Q(x_T) = 0$. For a localized instability (a bound state), the wave is trapped in a “potential well” between a left turning point ($-x_L$) and a right turning point (x_R).

According to WKB theory, for a bound state to exist and not destroy itself via destructive interference, the total phase accumulated by the wave as it travels from the left turning point to the right turning point must be a half-integer multiple of π . This gives the **Bohr-Sommerfeld Quantization Condition**:

$$\int_{-x_L}^{x_R} \sqrt{Q(x)} dx = \left(n + \frac{1}{2}\right) \pi, \quad n = 0, 1, 2, \dots \quad (16)$$

where n is the mode number ($n = 0$ is the fundamental mode, which usually has the highest instability growth rate).

2. Defining the Effective Momentum for Our System

From our previous derivation to $\mathcal{O}(v^2)$, the effective momentum squared $Q(x)$ is slightly different for $x > 0$ and $x < 0$ due to the Doppler shift of the background flow:

$$Q(x) = \begin{cases} Q_0^+ - |C_1|x^2 + C_2x^4 & \text{for } x > 0 \\ Q_0^- - |C_1|x^2 + C_2x^4 & \text{for } x < 0 \end{cases} \quad (17)$$

where the coefficients are:

$$|C_1| = \frac{\alpha^3 v^2}{2\omega^2} \quad (18)$$

$$C_2 = \frac{\alpha^2 v^2}{4} \quad (19)$$

The constant terms $Q_0^\pm$ contain the mean energy and a tiny discontinuous jump $\pm\delta$ at $x = 0$:

$$Q_0^\pm = Q_{0,\text{mean}} \pm \delta \quad (20)$$

$$Q_{0,\text{mean}} = \omega^2 - k_z^2 - \frac{\alpha^2 v k_z}{\omega^2}, \quad \delta = \frac{\alpha^2 v^2}{\omega} \left(1 - \frac{k_z^2}{\omega^2}\right) \quad (21)$$

Because of the discontinuity at $x = 0$, we must split the WKB integral into two halves:

$$\int_{-x_L}^0 \sqrt{Q_0^- - |C_1|x^2 + C_2x^4} dx + \int_0^{x_R} \sqrt{Q_0^+ - |C_1|x^2 + C_2x^4} dx = \left(n + \frac{1}{2}\right) \pi \quad (22)$$

3. The Perturbative Expansion of the Square Root

The integral of a quartic polynomial inside a square root produces Elliptic Integrals, which do not have simple algebraic forms. To obtain a closed-form dispersion relation, we assume the mode is highly localized near the shear layer $x = 0$.

In this inner region, the harmonic term $|C_1|x^2$ dominates the dynamics, and the quartic term C_2x^4 is a small perturbation. We can therefore expand the square root using the generic Taylor series $(1 - \epsilon)^{1/2} \approx 1 - \frac{\epsilon}{2}$:

$$\sqrt{A - Bx^2 + Cx^4} = \sqrt{A - Bx^2} \left(1 + \frac{Cx^4}{A - Bx^2}\right)^{1/2} \approx \sqrt{A - Bx^2} + \frac{Cx^4}{2\sqrt{A - Bx^2}} \quad (23)$$

Applying this to our integrals, the dominant leading-order physics of the instability is captured entirely by the unperturbed harmonic core:

$$\int \sqrt{Q_0^\pm - |C_1|x^2} dx \quad (24)$$

The quartic term provides minor shifting to the eigenvalues, but to track the fundamental onset of the complex roots analytically, evaluating the harmonic core is sufficient and rigorous.

4. Evaluating the Harmonic Integral

Let us evaluate the integral for the positive domain (from $x = 0$ to x_R). The turning point x_R occurs where the effective momentum goes to zero: $Q_0^+ - |C_1|x_R^2 = 0 \implies x_R = \sqrt{Q_0^+ / |C_1|}$.

$$I_+ = \int_0^{\sqrt{Q_0^+ / |C_1|}} \sqrt{Q_0^+ - |C_1|x^2} dx \quad (25)$$

To solve this, we use the trigonometric substitution:

$$x = \sqrt{\frac{Q_0^+}{|C_1|}} \sin \theta \implies dx = \sqrt{\frac{Q_0^+}{|C_1|}} \cos \theta d\theta \quad (26)$$

The integration limits change from $x \in [0, x_R]$ to $\theta \in [0, \pi/2]$. Substituting these into the integral:

$$I_+ = \int_0^{\pi/2} \sqrt{Q_0^+ - Q_0^+ \sin^2 \theta} \left(\sqrt{\frac{Q_0^+}{|C_1|}} \cos \theta \right) d\theta \quad (27)$$

Using the identity $1 - \sin^2 \theta = \cos^2 \theta$:

$$I_+ = \frac{Q_0^+}{\sqrt{|C_1|}} \int_0^{\pi/2} \cos^2 \theta d\theta \quad (28)$$

Since $\int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4}$, the exact value of the right half-integral is:

$$I_+ = \frac{\pi Q_0^+}{4\sqrt{|C_1|}} \quad (29)$$

By identical logic, the left half-integral is $I_- = \frac{\pi Q_0^-}{4\sqrt{|C_1|}}$.

5. Synthesizing the Final Dispersion Equation

Returning to the full split WKB condition (Equation 22), we sum the two halves:

$$\frac{\pi Q_0^-}{4\sqrt{|C_1|}} + \frac{\pi Q_0^+}{4\sqrt{|C_1|}} = \left(n + \frac{1}{2}\right) \pi \quad (30)$$

Substitute $Q_0^\pm = Q_{0,\text{mean}} \pm \delta$:

$$\frac{\pi(Q_{0,\text{mean}} - \delta)}{4\sqrt{|C_1|}} + \frac{\pi(Q_{0,\text{mean}} + \delta)}{4\sqrt{|C_1|}} = \left(n + \frac{1}{2}\right) \pi \quad (31)$$

The mathematical discontinuity $\pm\delta$ caused by the Doppler shift elegantly cancels out, leaving only the symmetric mean:

$$\frac{2\pi Q_{0,\text{mean}}}{4\sqrt{|C_1|}} = \left(n + \frac{1}{2}\right) \pi \implies \frac{Q_{0,\text{mean}}}{2\sqrt{|C_1|}} = n + \frac{1}{2} \quad (32)$$

Clearing the fraction gives the exact quantization condition for the harmonic core:

$$Q_{0,\text{mean}} = (2n + 1)\sqrt{|C_1|} \quad (33)$$